

Stakeholder's Requirements (Questions)

Our application allows Retailers to order supplies online. We are continuously working on optimizing the app.

- 1. Which KPIs would you use to measure the performance of our app?**

- 2. Prepare a report regarding our growth between the 2 years. Please try to answer the following questions:**
 - a. Did our business grow?
 - b. Does our app perform better now?
 - c. Did our user base grow?

- 3. What are our top-selling products in each of the two years? Can you draw some insight from this?**

- 4. Looking at July 2021 data, what do you think is our biggest problem, and how would you recommend fixing it?**

- 5. Does the login frequency affect the number of orders made?**

Business Insights

Q1. KPI's to Measure the Application Performance.

I am using some important KPIs to measure application performance.

Answer - To evaluate the overall performance of the application, I selected six key performance indicators that measure business growth, user activity, operational efficiency, and customer conversion. These KPIs provide a complete view of how effectively the application is performing across both business and user engagement dimensions.

1. **Total Orders** - Total orders in 2021 and 2022.

The total number of orders remained almost stable, with a slight decline of around **618 orders** in 2022. This indicates that order volume did not significantly grow, suggesting that the platform did not expand substantially in terms of transaction count.

However, order volume alone is not enough to judge app performance because fewer orders can still generate higher revenue if order value improves.

2. **Total Revenue** - Overall revenue for 2021 and 2022.

Revenue increased significantly in 2022. This is a major positive signal because despite slightly fewer orders, the platform generated much higher revenue.

3. **Total Users** - Number of users.

This represents the total customer pool available to the application.

4. **Active Users** - Number of Active users.

Active users declined significantly. This indicates a reduction in active engagement.

Possible reasons:

- User churn
- Reduced retention
- Lower repeat activity

This is a concern because a shrinking active user base can impact long-term growth.

5. **Logins** - Users who log in to the application.

Login activity remained almost identical. This is interesting because although active users decreased, total logins remained stable.

This suggests: **Fewer users are generating similar login activity**

Meaning: The remaining active users are engaging more frequently with the app. This indicates stronger engagement among retained users.

6. **Order Status** - Status of orders (Shipped, rejected, review, pending)

The number of rejected orders is significantly higher than shipped orders. This is a critical operational issue.

It suggests:

- Inventory shortages
- Fulfillment bottlenecks
- Process inefficiencies

High rejection directly affects:

- Customer trust
- Revenue realization
- User retention

This is one of the most important problem areas identified in the analysis.

Q2. Report regarding our growth between the 2 years

Did Our Business Grow?

To measure business growth, I will analyze key metrics such as

- a. Total Number of Orders
- b. Number of successfully completed orders
- c. Total revenue generated between 2021 and 2022

Answer - I analyzed the growth between 2 years, and the successful orders are fewer than the rejected orders, although the revenue is higher in 2022 compared to 2021

Does our app perform better now?

To measure the application performance, I will analyze key metrics such as

- a. Order Success Rate
- b. Conversion Rate

Answer - To evaluate app performance, I analyzed conversion rate and order success rate. The conversion rate increased from 18% in 2021 to 24% in 2022, indicating that more users are placing orders after logging in. Additionally, the order success rate improved from 39% to 48%, showing better order fulfillment efficiency. Based on these improvements, the application performance has clearly improved in 2022.

Did our User Base grow?

To measure the user base growth, I will analyze key metrics such as

- a. Number of active users.

Answer - Although the number of active users decreased in 2022, both conversion rate and order success rate improved significantly. This indicates that while the user base shrank, the quality and engagement of users improved, leading to better overall performance.

Q3. What are our top-selling products in each of the two years? Can you draw some insight from this?

Answer - The top-selling product in 2021 was Product number **10235**, while in 2022 it changed to Product number **12547**. This indicates a shift in customer preferences, possibly due to changes in demand, pricing, or product availability. Additionally, the quantity sold increased from 1084 to 1560, suggesting that the new top product has a stronger demand and better market acceptance.

Q4. Looking at July 2021 data, what do you think is our biggest problem, and how would you recommend fixing it?

Key Metrics:

- a. Order Status Distribution
- b. Conversion rate
- c. Success rate (shipped/total orders)

Answer - Looking at July 2021 data, the biggest problem is the high order rejection rate. Out of total orders, a significantly higher number of orders were rejected (4140) compared to successfully shipped orders (2655). This indicates a major issue in order fulfillment, such as inventory shortages, supply chain inefficiencies, or operational delays. To address this, I would recommend improving inventory management, ensuring better stock availability, and optimizing the order processing system. Reducing rejection rates would directly improve customer satisfaction, conversion rates, and overall business performance.

Q5. Does the login frequency affect the number of orders made?

Answer - There is a moderate positive relationship between login frequency and the number of orders, indicating that users who log in more frequently tend to place more orders. However, a significant number of users have high login activity but zero orders, suggesting a gap between user engagement and conversion. This indicates that while users are interacting with the app, they are not being effectively converted into customers.